TRIBHUVAN UNIVERSITY

Institute of Science and Technology



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partial fulfillment of the requirement for the Master's degree in Information Technology

CALCULATION OF EJECTION FRACTION OF ECHOCARDIOGRAPHY USING DEEP LEARNING ALGORITHM

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Tribhuvan University
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Supervisor Recommendation

This is to certify that Mr. Sujan Kafle has submitted the Dissertation report on the topic "**Calculation of Ejection Fraction of Echocardiography Using Deep Learning Algorithm**" for the partial fulfilment of Masters of Information Technology, Third semester. I hereby, declare that this Dissertation report has been approved.

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Approval Letter

This is to certify that the Dissertation report prepared by Mr. Sujan Kafle on the topic" **Calculation of Ejection Fraction of Echo Cardiography Using Deep Learning Algorithm**" in partial fulfilment of the requirements for the degree of Masters of information Technology has been well studied. In our opinion, it is satisfactory in the scope and quality as a project for the required degree.

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Thanking you,
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Plagiarism Test Report

ABSTRACT

In order to determine the ejection fraction (EF), a crucial measure of heart function, this study investigates the use of supervised learning (SL) for segmenting and evaluating echocardiogram pictures. For accurate semantic segmentation, the work incorporates robust feature representations that were learnt from labelled echocardiogram data into a U-Net architecture. By providing efficient pre-training on labelled data, this method tackles typical issues in medical imaging, such as restricted access to annotated datasets because of high costs and privacy restrictions. The suggested approach achieved an average IoU of 0.8967 and a Dice score of 0.9102 for segmenting the regions of the heart when assessed using Intersection over Union (IoU) and Dice Coefficient measures. End-diastolic and end-systolic volumes were calculated using the segmented sections to calculate the EF, enabling precise and automated cardiac evaluation. The outcomes show significant gains over conventional techniques, underscoring SL's promise in medical imaging applications. This work offers a scalable framework for automated cardiac diagnostics by combining segmentation with effective feature extraction. This reduces dependency on huge labelled datasets while improving model accuracy and generalization.

Keywords: Reset, U Net3++, Segmentation, Echocardiography

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List of Abbreviations

A4C	Apical Four Chamber
ACC	American College of Cardiology AI Artificial Intelligence
CAMUS	Cardiac Acquisitions for Multi-Structure Ultrasound Segmentation
CNN	Convolutional Neural Network
EDV	End Diastolic Volume
EF	Ejection Fraction
ESV	End Systolic Volume
GCN	Graph Convolutional Networks
IoU	Intersection over Union
LV	Left Ventricle
LVEF	Left Ventricular Ejection Fraction
ML	Machine Learning
ReLU	Rectified Linear Unit
ROI	Region of Interest
SL	Supervised Learning
SV	Stroke volume

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Unit 1: Introduction

1.1. Introduction

Heart and blood vessel illnesses are grouped together as cardiovascular diseases (CVDs). Rheumatic heart disease, peripheral artery disease, cerebrovascular disease, and coronary heart disease are among them. Deep vein thrombosis, pulmonary embolism, and congenital heart disease. Diagnostic imaging is an invaluable tool in medicine today [1].

The percentage of blood that is pumped out of the left ventricle of the heart with each heartbeat is known as the Ejection Fraction (EF). It is a crucial sign of heart health. The heart pumps out more than half of the blood it contains with each beat when the EF is between 55% and 70%. A preserved EF can still occur in some forms of heart failure, while a low EF (below 40%) may indicate the heart isn't pumping blood effectively. EF is used to diagnose and track heart conditions, particularly heart failure.

Digital mammography, computed tomography (CT), magnetic resonance imaging (MRI), and other imaging modalities offer a useful way to map a subject's anatomy without intrusive procedures. These technologies are essential for diagnosis and therapy planning, and they have significantly expanded our understanding of both healthy and sick anatomy for medical study. Computers have become essential for processing and analyzing medical images due to their growing size and quantity. Specifically, computer algorithms that help and automate certain radiological processes are crucial for the identification of anatomical structures and other locations of interest. In many biomedical imaging applications, including tissue volume quantification, diagnosis, pathology localization, anatomical structure study, treatment planning, partial volume correction of functional imaging data, and computer-integrated surgery, these algorithms—known as image segmentation algorithms—are essential [2]. Depending on the particular application, imaging modality, and other variables, there are considerable variations in segmentation techniques. For example, because of its anatomy, the brain is segmented differently from the liver.

1.2. Problem Statement

Ejection fraction is a critical biomedicine metric used to assess heart function, particularly for diagnosing heart failure. Cardiologists' manual estimation takes a lot of time and is prone to inter-observer variability. The main aim is to use segmentation-based architectures (U-Net) to accurately predict EF from 2D echocardiogram frames or videos, reducing workload and improving consistency.

1.3. Objectives

1. To utilize Deep learning algorithms for calculated the ejection fraction (EF) using the Segmented region of the heart and to determine if it needs for medical support or not.
2. To implement Supervised learning and improves computational efficiency and speed by reducing the reliance on large labeled datasets during segmentation tasks.

1.4. Scope and Limitations

Supervise learning has significant potential in medical image segmentation research, where labeled data is scarce and large-scale unlabeled datasets are abundant. It leverages vast amounts of unlabeled medical images to learn useful feature representations to reduces the need for large-scale annotated datasets, a common challenge in medical imaging [3][4]. Pre-trained contrastive encoders can be fine-tuned for segmentation tasks, improving accuracy and efficiency. Also, particularly useful for organ or tumor segmentation in modalities like MRI, CT, and ultrasound. It employs augmentation techniques to create positive and negative pairs, enriching the dataset's diversity without additional labeling.

Supervise learning has inherent challenges and limitations too. Designing domain specific augmentations is critical but difficult for medical images, as inappropriate augmentations may distort diagnostically important features from medical images. It is unsupervised technique and not make to segmentation tasks, which may result in suboptimal representations for delineating structures or regions. It requires large-scale unlabeled datasets to learn meaningful representations, which might not always be available in specific medical domains. While training contrastive models, it is computationally expensive, requiring large batch sizes or memory banks to maintain a wide pool of negative samples. While it reduces annotation needs during pretraining, labeled data is still required to evaluate segmentation performance and for downstream fine-tuning.

1.5. Report Organization

The Report organization of this thesis is as follows:

Unit 1 Introduction: The background and purpose of the study are explained in the introduction section. It explains the issue being tackled, the study's goals, the extent of the work, and the constraints faced during the investigation.

Unit 2 Background Study and Literature Review: The theoretical ideas, important research, and current techniques pertinent to the subject are described in the background study and

literature review section. It explains the gaps in the literature and explains why the suggested method is necessary.

Unit 3 Methodology: The general research strategy is explained in the methodology section. The dataset, data pretreatment procedures, models or algorithms employed, and performance evaluation measures used to evaluate the model are all described.

Unit 4 Implementation: The practical application of the research is explained in the implementation section. It outlines the methods, instruments, and procedures used to prepare the data, train the model, and carry out testing and validation.

Unit 5 Result and analysis: The findings of the experiments are described in the results and analysis section. It explains crucial findings, quantitative results, visual outputs, and how the model's performance was interpreted.

Unit 6 Conclusion and Future Recommendations: The study's general conclusions are presented in the conclusion and future recommendations section. It outlines the key contributions, draws attention to the drawbacks, and offers recommendations for future enhancements and additional study.

Unit 2: Background Study and Literature Review

2.1. Background Study

A crucial clinical metric for evaluating the left ventricle's pumping efficiency in echocardiography, the ejection fraction (EF) is essential for both identifying and treating cardiovascular disorders. EF measurement has relied on the laborious, [5] operator-dependent, and human error-prone manual drawing of ventricular boundaries from echocardiographic pictures. Automated EF calculation has become a viable way to increase accuracy, efficiency, and repeatability with the development of artificial intelligence, especially deep learning. With little assistance from humans, deep learning algorithms can automatically segment the left ventricle and estimate EF from echocardiographic pictures. Despite these developments, issues including clinical validation, data variety, and picture quality variance still exist. Thus, creating a strong deep learning-based model for automatic EF estimate should greatly improve diagnostic accuracy and increase accessibility to cardiac evaluation, particularly in healthcare settings with low resources.

2.2. Literature Review

In this paper, [6] authors used to find accurate assessment of cardiac function and diagnosis of cardiovascular disease using deep learning algorithm. They used algorithm for segmenting left ventricle, estimating ejection fraction and classifying cardiomyopathy disease. For frame-level semantic segmentation of the left ventricle, they employed a CNN model with atrous convolutions, and for ejection fraction prediction, they employed a CNN model with residual connections and spatiotemporal convolutions across frames. Their model identifies heart failure with reduced ejection fraction (AUC of 0.97), predicts ejection fraction with mean absolute error of 4.1%, and segments the left ventricle with a Dice Similarity Coefficient of 0.92. In order to calculate clinical indices like ventricular volume and ejection fraction, the left ventricle (LV) must be segmented from cardiac magnetic resonance imaging (MRI) datasets. The authors of this work developed and assessed a fully automatic segmentation tool for the left ventricle using short-axis cardiac MRI datasets using deep learning techniques in conjunction with deformable models [7]. This method provides high accuracy only with 2D dataset, and will not work with 3D MRI dataset and lack of sufficient dataset for train and validation purpose. In this paper [8], author generate automated method called Echo Graphs for predicting ejection fraction and segmenting the left ventricle by detecting anatomical key points with Graph Convolutional Networks (GCNs). The problem with the research was, low image quality and false annotation

in the dataset. In this paper [9], AI approach for deriving advanced biomarkers of systolic and diastolic LV function from 2-D echocardiography based on segmentations of the full cardiac cycle [4] and this AI model is based on the 'UNet' framework. Due to low frame rate i.e. (>=20 frames/second) will cause model difficulties in estimating the volume derivatives to compute systolic/diastolic biomarkers. In this paper [4], author introduce state-of-the-art encoder-decoder deep convolutional neural network methods can go at assessing 2D echocardiographic images i.e. segmenting cardiac structures as well as estimating clinical indices. The Cardiac Acquisitions for Multi-Structure Ultrasound Segmentation (CAMUS) dataset used for the purpose of echocardiographic assessment. Paper show that encoder-decoder based architectures outperform state-of-the-art non-deep learning methods and faithfully repro duce the expert analysis for the end-diastolic and end-systolic left ventricular volumes [10] [11]. The problem with the research is, poor contrast 2D echocardiographic images due to 3 reverberations, clutter, acoustic shadowing in some regions of the images. In this paper [12], a real-time and fully automated EF (ejection fraction) measurement and foreshortening detection method is proposed and uses several deep learning components like view classification, cardiac cycle timing, segmentation and landmark extraction, to measure the amount of foreshortening, LV volume, and EF. EF is considerably affected by apical foreshortening; proposed LV length difference and longitudinal apex movement metrics have the potential of detecting and quantifying apical foreshortening.

The paper [13] proposed method that is based on multi-planar deep convolutional neural networks (CNN) with an adaptive fusion strategy where authors automatically utilize complementary information from different planes of the 3D scans for improved delineations. They have independently designed three CNNs (with the same architectural configuration) for three planes of CT and MRI images. They have trained the networks from scratch for voxel-wise labeling for the following cardiac structures: ascending aorta (Ao), main pulmonary artery (PA), left atrium (LA), left ventricle (LV), right atrium (RA), right ventricle (RV), and myocardium of the left ventricle. The research is hindered by the usage of SoftMax function in the final layer of the network and the lack of a large number of 3D images to execute the 3D CNN approach. Deep neural networks have shown a great potential in image pattern recognition and segmentation for a variety of tasks. In this paper [14], author demonstrate an auto mated analysis method for CMR images, which is based on a fully convolutional network (FCN). The network is trained and evaluated on a large-scale dataset from the UK Biobank, consisting of 4,875 subjects with 93,500 pixelwise annotated images. Problem with this research is that, model is trained on single dataset which is homogeneous in nature of UK biobank dataset. In this paper [15], author propose a

supervised contrastive learning method to segment the left ventricle (LV) from echocardiography when limited annotated images exist. Also, they have investigated how two popular segmentation networks, DeepLabV3 and U-Net, are affected by contrastive pre-training. In this paper [16], When there is a lack of labeled data, contrastive pre-training aids in improving left ventricle segmentation ability.

Unit 3: Methodology

3.1. Description of Methodology

This research will perform experimentation methodology, by implementing a machine learning algorithm that will find representation or annotations of the images using Supervised learning (U-Net) with limited dataset.

Steps:

1. Dataset (Labeled Echo net Dynamic dataset).
2. Image extraction and preprocessing using Python and OpenCV.
3. Data preparation results in fully labeled datasets.
4. Train a U-Net segmentation model using the labeled data.
5. Evaluate model performance and output results

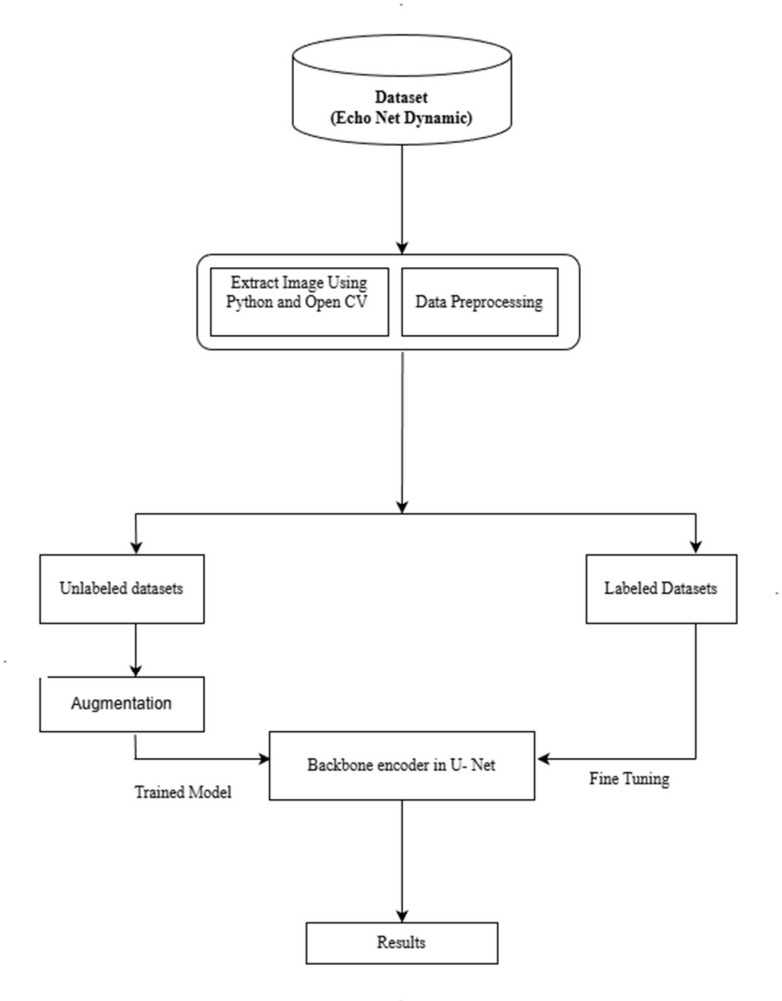


Figure 3 1:Proposed Block Diagram

3.2. Data Set Description

3.2.1: EchoNet-Dynamic

The dataset was collected from EchoNet-Dynamic, an open-source dataset made available to the general public by Stanford University. The dataset was created primarily for tasks like segmentation, left ventricular ejection fraction estimation, and other evaluations of cardiac function in deep learning research related to cardiac imaging. 10,030 annotated echocardiography films in the four-chamber view of the heart are included, together with expert human labels to create a reference for examining chamber size and cardiac action. Patients at Stanford University Hospital provided the data between 2016 and 2018. Two cardiac ES and ED frames are individually draught and labelled for each video. Each video in my dataset shows the structure of the left ventricle at two different times: end-diastole and end-systole. Expert annotators trace the ventricle's endocardial border throughout each phase of the film. Ventricular volume is computed using these tracings by integrating the area they contain along the primary axis of the ventricle. A set of paired coordinates is used to represent the tracings. The length and orientation of the ventricle's long axis are indicated by the first pair, and the lengths from the heart's apex to particular spots along the ventricle's short axis are indicated by the following pairs. Every video coordinate pair is linked to the precise frame of the video where the tracing is done, along with the file name and frame number of each video file. To the best of our knowledge, this is one of the largest publicly available datasets for left heart component segmentation. Because it contains a significant amount of both annotated and unannotated data, it is very helpful in achieving our study goal.

3.3. Data Preprocessing

3.3.1 Dataset preparation

10,030 datasets were collected from Standford University Publicly available dataset. The data preparation was carried out as follows:

Step 1: Pictures were taken from videos in which there were 20 frames per segment.

Step 2: To extract frames from all 10,030 echocardiography videos, a Python script was created.

Step 3: Using the tracing information supplied, videos were divided into separate frames.

Step 4: Frames were extracted and image sequences were created using the MPEG image sequence tool.

Step 5: To extract frame-by-frame images, OpenCV Video Capture was employed as an additional technique.

Step 6: To preserve the extracted frames' original temporal order, they were rearranged.

Step 7: Images were divided into masked (for segmentation) and unmasked (for U-Net pre-training) sets.

Step 8: Pre-training and segmentation folders with train, test, and validation subfolders were created from the final dataset.

Table 3. 1.Dataset of Echocardiography

S.no	Data Size	Purpose
1	Train (116917), Test (20360)	Unlabeled Dataset
2	Train (14920), Test (2552)	Labeled Dataset

dataset:



Figure 3 2:Unlabeled Image of Echocardiography



Figure 3 3: Labeled Image of Echocardiography

3.3.2. Data Augmentation

Supervised learning methods were used in the study using labelled datasets. Each dataset was first subjected to data augmentation in order to create two versions of each image for use in supervised learning in order to satisfy the requirements. A number of augmentation techniques were used, such as [you may enumerate the techniques here.

• **Resizing Image:** Sometime the input images are too big to process, so we have to resize them so that our model can trained effectively. We need to resize all images, including those generated through augmentation, because convolutional neural networks require input data of uniform dimensions for training. The original image size was 112 x 112, so we performed resizing to achieve an augmented image size of 224 x 224

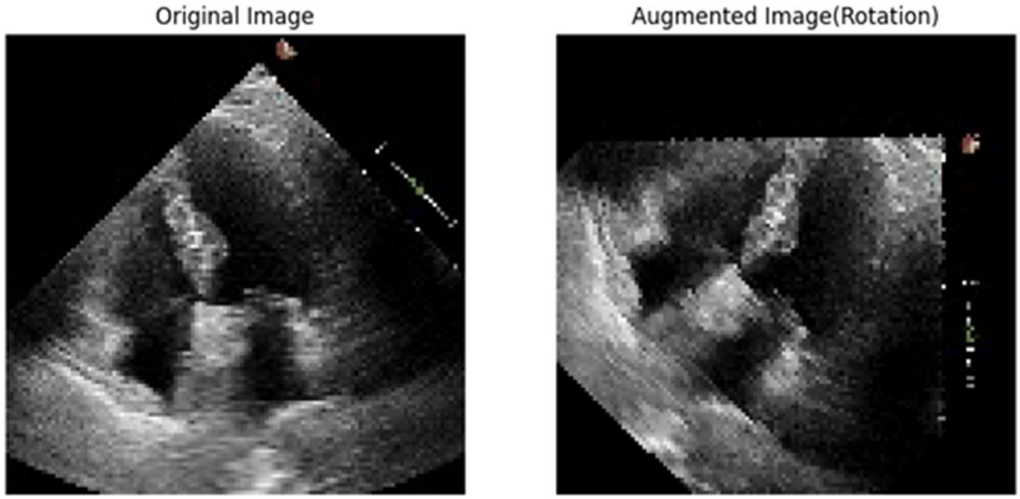


Figure 3 4:Rotation Augmentation Image

Horizontal Flipping: In this technique, the image can be flipped horizontally in a random manner based on a specified probability, p. If p equals 1, a horizontally f lipped image is always returned. If p equals 0, the original image is retained When p lies between 0 and 1, the likelihood of obtaining a horizontally flipped image corresponds to the value of p.

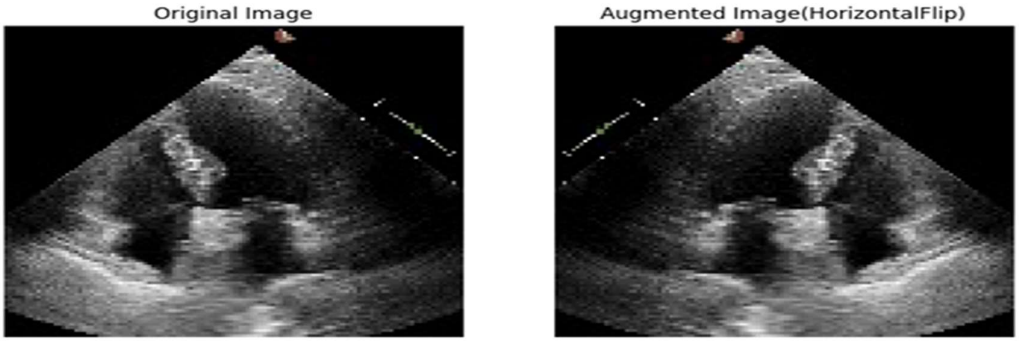


Figure 3 5: Horizontal Flip Augmentation Image

Vertical Flipping: This is another method of flipping an image. In this case, the image is flipped vertically at random with a specified probability value, p. When p=1, it always produces a vertically flipped image. When p=0, it retains the original image. For p values in the range (0,1), the likelihood of obtaining a vertically flipped image corresponds to p

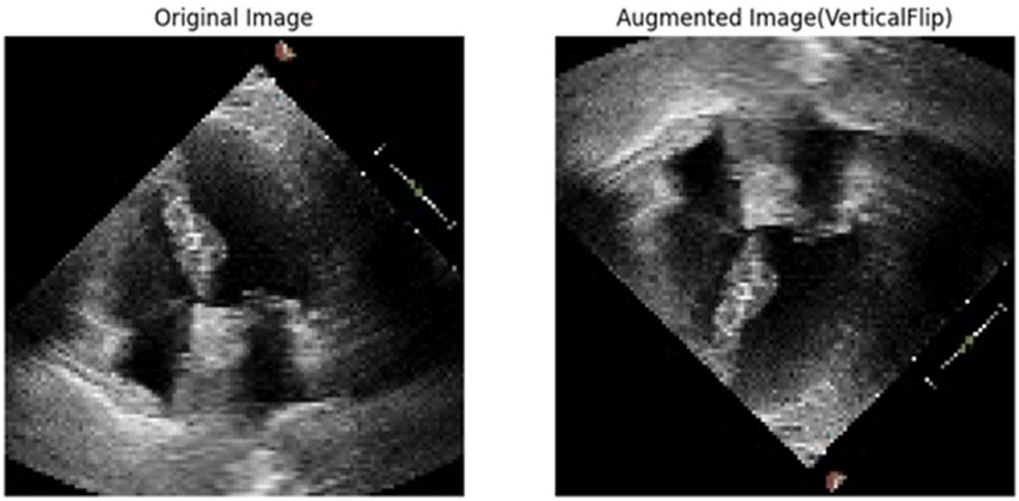


Figure 3 6: Vertical Flip Augmentation Image

Color Transformation: This is another augmentation method that is used to get different variety of images. Generally, color transformation uses color transformation level to change the color of the input image. Here we have used brightness in range (0.1,0.6), contrast=1, saturation=1 and hue=0.4.

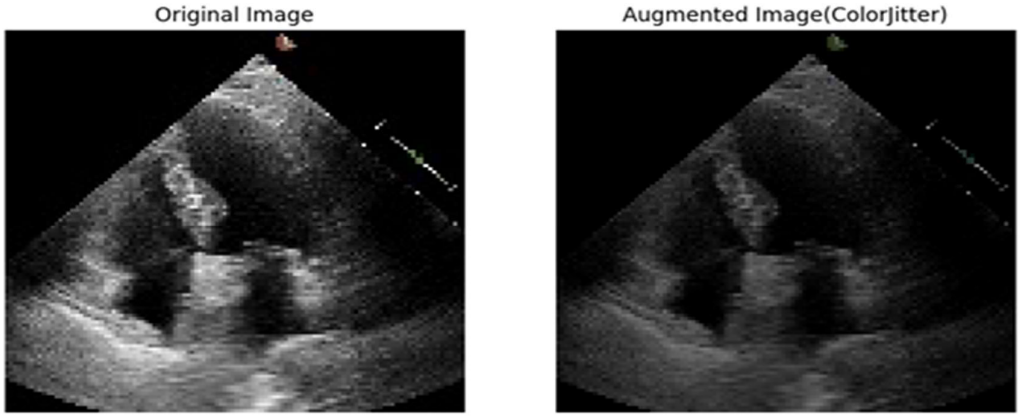


Figure 3 7: Color Jitter Augmentation Image

Random Perspective: In this augmentation technique, we use it to random scaling perspective transformation of image. In this research we have used distortion scale value of 0.5, p=0.5 and fill=0 value to get random perspective augmentation of input image.

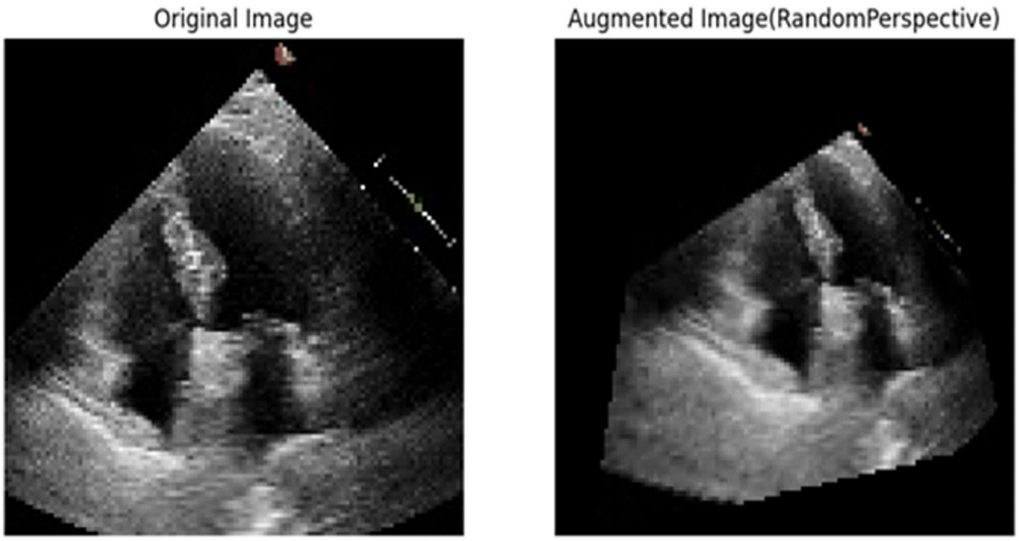


Figure 3 8: Random Perspective Augmentation Image

3.4. Description of Algorithms/ Mathematical Models

3.4.1 Resnet

An ordinary method for resolution of complicated challenges with the use of neural net- works is by adding of some more extra layers to the model, making the model deeper. By which, we can increase model accuracy helps to learn more features from input in- formation. Nevertheless, there is a problem of vanishing or exploding gradients, which means that the error rate of the model also increases [17]. To solve this issue, a new neural architecture called Residual Network, or ResNet, was injected . ResNet, can be used as a backbone architecture structure of an encoder model. This means that resnet is the network that is used to extract the learned features representation from the input data, and in which way, [18] ResNet is used in our research work as an encoder input to implement supervised contrastive learning architecture model.

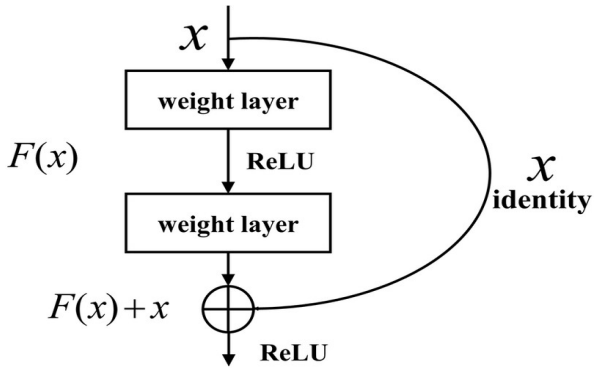


Figure 3 9: Building Block of ResNet

Pseudocode of Resnet

function ResidualBlock(x):

y = Conv3x3(x)

y = BatchNorm(y)

y = ReLU(y)

y = Conv3x3(y)

y = BatchNorm(y)

out = ReLU(x + y)

return out

3.4.2. Unet

The u-net is a convolutional network architecture designed to segment images quickly and accurately as shown in Figure 3.10. As of right now, it has surpassed the previous best technique for picture segmentation on the ISBI challenge for segmenting neural structures in electron microscopic stacks, also referred to as a sliding window convolutional network method [19].

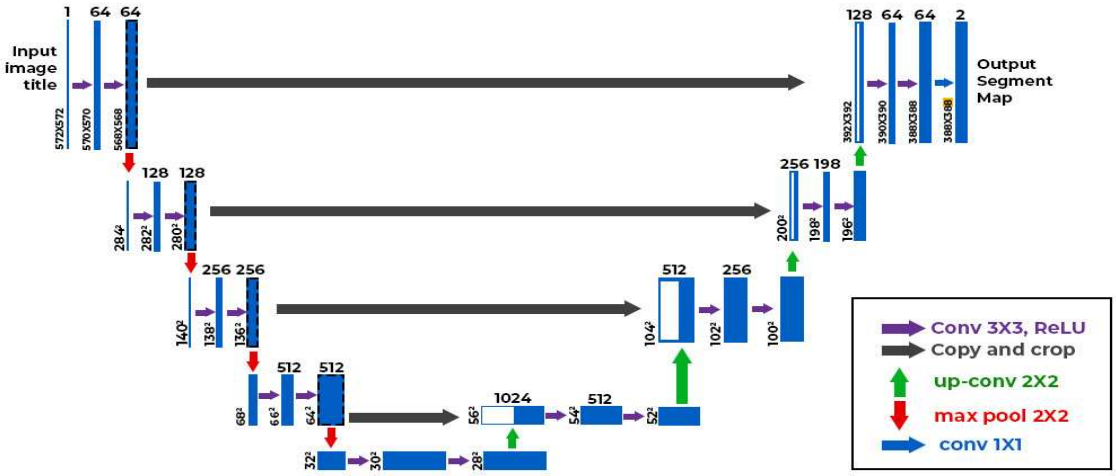


Figure 3 10: U-Net Architecture

There are two primary paths in the architecture: an expansive path and a contracting approach. Similar to traditional convolutional neural networks, the contracting path consists of two 3x3 convolutional layers applied repeatedly without padding, each followed by a ReLU activation function. The number of feature channels is doubled at each down sampling step and spatial dimensions are reduced using a 2x2 max pooling procedure with a stride of 2. The extended approach allows for accurate localization and reconstruction of the segmented output by up sampling high-level feature maps and then reducing the number of feature channels by a 2x2 convolution.

After merging the segmentation output with the matching cropped feature map from the contraction path, it is processed through two consecutive 3x3 convolutional layers and then a ReLU activation function. In order to make up for the loss of boundary pixels during convolution operations, feature maps must be cropped. At the end of the expanding route, the segmented output with dimensions of 224x224 is produced by applying a 1x1 convolution to map each 64-dimensional feature vector to the desired number of output classes. As shown in Figure 3.4, the entire network consists of 23 convolutional layers and is intended to handle input images with dimensions of 3×224×224. This architecture guarantees accurate segmentation of the input echocardiogram pictures and exact feature extraction.

Pseudocode of U Net

Encoder Block

x1 = Down(x)

x2 = Down(x1)

x3 = Down(x2)

x4 = Down(x3)

Bottleneck
b = ConvBlock (x4)

Decoder with skip connections

u3 = Up(b, x3)

u2 = Up(u3, x2)

u1 = Up(u2, x1)

u0 = Up(u1, x)

Output
out = Conv1x1(u0)

Mathematically

Decoder Block

$$\left(W_i^{(4)} * [\,\sigma\left(W_i^{(3)} * (\,\mathrm{UpConv}(D_{i+1}) \;\oplus\; E_i\,)\right)]\right) \quad (1)$$

Where,

D_{i+1} = output from the next (deeper) decoder layer

$\mathrm{UpConv}(\cdot)$ = upsampling (transpose convolution)

$\oplus$ = concatenation with encoder skip feature

E_i = encoder feature from same level (skip connection)

$W_i^{(3)}, W_i^{(4)}$ = convolution weights in decoder layer i

σ = activation function

D_i = output feature map of decoder layer i

Encoder Block:

$$\left(W_i^{(2)} * [\,\sigma\left(W_i^{(1)} * E_{i-1}\right)]\right) \quad (2)$$

Where,

E_{i-1} = input to encoder layer i

$W_i^{(1)}, W_i^{(2)}$ = convolution weights in encoder layer i

σ = activation function

E_i = output feature map of encoder layer i

Output Layer:

$$\hat{Y} = W_o * D_1 + b_o \quad (3)$$

Where,

D_1 = output from the final decoder block

W_o, b_o = final convolution weights and bias

$\widehat{Y}$ = segmentation l (apply sigmoid/SoftMax afterward)

3.4.3. Segmentation

Image segmentation is a operation in image vision where the objective of it is to group each and every pixel of image into one more group or class. As comparison with image classification, where the objective of it is to group the image as a whole bundle, which is highly hard to perform. Among them one of the difficulties of semantic segmentation is that it must have more understanding of the semantics of the images to perform better and provide good result. For segmentation purpose we use U-Net architecture [21], where we use our previously trained contrastive learned features as backbone encoder to our classic U-Net architecture.

After applying pre-trained model encoder into U-Net as backbone encoder [47]. Then fine-tuned the model with using some labeled or mask dataset. Which will help segmentation model to work perfectly. We used U-Net model with labeled dataset for fine tuning model so that it will segment properly with the help of the contrastive learned features.

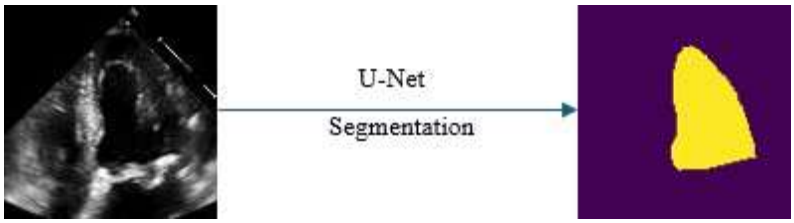


Figure 3 11: Segmentation

3.4.4. Ejection Fraction

It is the main analytic method used in the early phase analysis of the heart failure. It is computed as the proportion of the stroke volume of blood to the end diastolic volume which is also the percentage of blood pumped from the left ventricle of the human heart. The end-diastole area is the largest area of the heart part at the point when the heart is filled with blood, and the end-systole area is the smallest area of the LV at the point A crucial measure

of heart health is the left ventricular ejection fraction (LVEF) [8][9], which is the ratio of the left ventricle's total volume at the conclusion of diastole to the amount of blood expelled from it during systole. The heart's stroke volume (SV) is determined by subtracting the end-diastolic volume (EDV) from the end-systolic volume (ESV). In this work, we use echocardiogram pictures to estimate the ejection fraction by examining the left ventricle's segmented sections. By adding up every pixel in the segmented region, the segmented area is determined. End-diastolic and end-systolic phases are represented by the maximum and minimum segmented areas, respectively. The ejection fraction rate is then determined using these measures.

Mathematically, LVEF is calculated as:

$$\text{LVEF}\ (\%) = \frac{\text{EDV} - \text{ESV}}{\text{EDV}} \times 100 \qquad (4)$$

where,
Stroke Volume (SV), End Diastolic Volume (EDV) and End Systolic Volume (ESV).
The simplest classification as per the ACC that is used clinically as follows:

Table 3. 2. Problems with Ejection Fraction rate

S.No.	Ejection Fraction	Classification	Description
1	>70 %	Hyperdynamic	Excessively high EF; typically not a concern unless symptomatic.
2	50-70%	Normal	Normal heart function.
3	40-49%	Mild dysfunction	Mild decrease in heart function; early heart failure may be present.
4	30-39%	Moderate dysfunction	Moderate heart failure; requires medical intervention.
5	<30%	Severe dysfunction	Severe heart failure; significant risk of complications.

1. Hyper dynamic: This is typically considered a high EF. While this can occur in healthy individuals, an unusually high EF might be seen in certain conditions such as hyperthyroidism or arteriovenous fistulas. However, it is generally not a cause for concern unless associated with other symptoms or conditions. In this range, the heart pumps blood more effectively than usual, but this could also reflect abnormal compensatory mechanisms in certain pathologies.

2. Normal: This is considered a normal range, meaning the heart is pumping effectively and efficiently. Individuals with an EF in this range typically do not have any major concerns related to heart function.

3. Mild Dysfunction: This range indicates mild left ventricular dysfunction. While not immediately life-threatening, it suggests that the heart is not pumping as effectively as it should, but there is still sufficient output for normal activities. People in this range might have early-stage heart failure, especially if other symptoms (like shortness of breath, fatigue, or swelling) are present. It may also suggest the early stages of heart disease or damage due to past heart attacks.

4. Moderate Dysfunction: This indicates moderate dysfunction. The heart is significantly less efficient at pumping blood, which can lead to fluid buildup and symptoms of heart failure. People with an EF in this range may experience more noticeable symptoms such as fatigue, difficulty exercising, and shortness of breath. Patients might be at risk of heart failure and require medications or treatments to prevent further decline in heart function.

5. Severe Dysfunction: This is considered severe dysfunction, indicating that the heart is pumping less than 30% of its blood with each contraction. This is a criti- cal condition that often results in heart failure symptoms, such as fluid retention, swelling, and severe shortness of breath. Individuals with an EF below 30% are at high risk for severe complications.

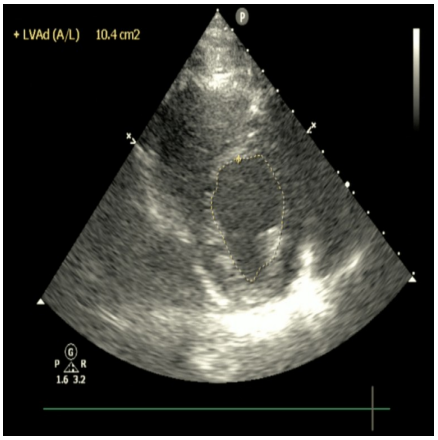


Figure 3 12: .Left Ventricular During Systolic



Figure 3 13: Left Ventricular During Systolic

3.5. Performance Evaluation Metric

For model evaluation, we used Intersection over Union (IoU) and Dice Similarity Coefficient (Dice-coefficient), these are evaluation metrics commonly used in image segmentation tasks to calculate accuracy of output segmentation masks compared to labeled masks.

3.5.1. Intersection Over Union

This evaluation metric evaluates degree of intersection of the output segmented value and real actual labeled value. It is calculated as proportion between intersection area of output value and real labeled value to total union covered value. [51].

Mathematically:

$$IoU = \frac{|A \cap B|}{|A \cup B|} \tag{5}$$

where |A ∩ B| is the intersection area and |A ∪ B| is the union area of the predicted mask A and the ground truth mask B.

3.5.2. Dice Coefficient

This evaluation technique calculates the similarity between two sets. While in image segmentation area, it determines how similar of the output value is to the labeled value. It is computed as twice of intersection of the output mask and labeled masks divided by the sum of areas of output mask and labeled mask. [52].

Mathematically:

$$\text{Dice} = \frac{2 * |A \cap B|}{|A| + |B|} \tag{6}$$

where $|A \cap B|$ is intersection area, $|A|$ is the area of the output mask image A, and $|B|$ is the area of the labeled image mask B.

Here both evaluation metrics values lie in the range 0 to 1. Here 0 stands for lower value and similarly 1 stand for higher value. More the toward the higher value means our model is performing better. These metrics are used for assessing the accuracy of segmentation model.

Unit 4: Implementation

4.1. Tools and Technique

4.1.1: Tools

To implement this research, following hardware and software configurations were used:

Hardware System Primary System – TUHPC (Supercomputer):

- Processor: Intel® Xeon® CPU E5-2680 v2 @ 2.80GHz
- RAM: 5 TB
- Storage: 60 TB
- Purpose: High-performance computing for training deep learning models remotely.
- GPU: 200 GB

Alternative System (Local Workstation):

- Processor: Intel Core i7 HX @ 3.5 GHz
- RAM: 24 GB DDR5
- Graphics: NVIDIA RTX 4060, 8 GB VRAM
- Storage: 512 GB NVMe SSD
- Purpose: Local experimentation and testing.

Software Configuration

- Operating System: Linux (Primary), Windows (for local workstation)
- Programming Language: Python 3.15
- Framework: PyTorch
- Package & Environment Management: Anaconda
- GPU Computing: CUDA 12.9 (for NVIDIA GPU acceleration)

Supporting Tools

- PuTTY: Remote login to TUHPC
- WinSCP: Secure file transfer between local system and supercomputer.

4.1.2. Techniques

To estimate Ejection Fraction from echocardiogram pictures, the study uses a number of techniques. In order to increase model generalization, data preprocessing involves standardizing images using pixel normalization, resizing them to a uniform resolution for neural network input, and enhancing them with data augmentation techniques including rotation, flipping, and scaling.

During the modelling stage, fully connected layers are used to predict continuous EF values after a U-Net architecture is used to extract features from echocardiogram pictures. The model employs learning rate scheduling to dynamically modify the learning rate, the Adam optimizer for gradient-based weight updates, and Binary Cross-Entropy (BCE) loss for supervised learning during training.

To maximize GPU memory use, batch training is used with 8 samples for training/validation and 1 for testing. Lastly, the Anaconda environment guarantees consistency and usability across systems, and Python scripts automate EF prediction on fresh images for deployment.

4.2. Implementation Details

4.2.1. Image Extraction

A large number of echocardiography videos continuous recordings of heart motion were gathered for this study's first phase. Since raw video files are not directly appropriate for model training, the videos were edited to extract individual image frames that captured important cardiac features at different time points. The unstructured video data was converted into a well-organized picture dataset by this frame extraction step, which serves as the foundation for the preprocessing, segmentation, and analysis processes that follow. This approach to data preparation guarantees that all subsequent phases of the research are founded on precise, trustworthy.



Figure 4. 1.Image Extraction from video of eco-cardiography.

4.2.2. Data Argumentation

To improve the model's resilience during training, data augmentation is used to echocardiogram pictures. Images and their matching masks are loaded from designated directories by the dataset class Echocardiography Dataset. The photographs are enhanced through a variety of modifications using the Albumentations library. These include shifting, scaling, rotation, and random horizontal and vertical flips to replicate various heart orientations, as well as random brightness and contrast alterations to replicate changes in image

quality. Furthermore, blur is used to mimic minor distortions from imaging equipment, and Gaussian noise is used to mimic frequent ultrasound image artefacts.



Figure 4. 2. Data Argumentation Working Process.

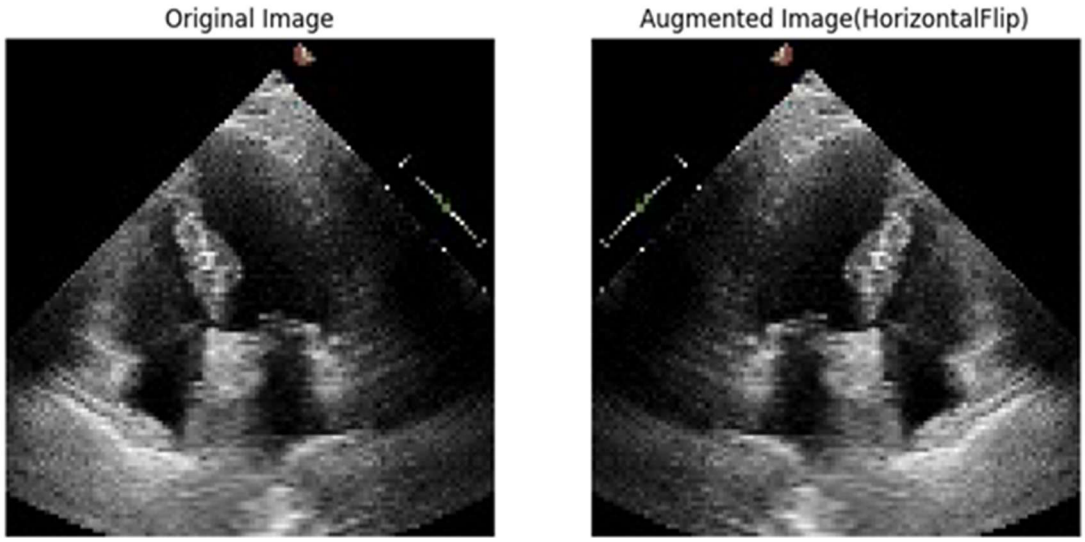


Figure 4. 3. Horizontal Flipping

If p equals 1, a horizontally flipped image is always returned. If p equals 0, the original image is retained When p lies between 0 and 1, the likelihood of obtaining a horizontally flipped image corresponds to the value of p.

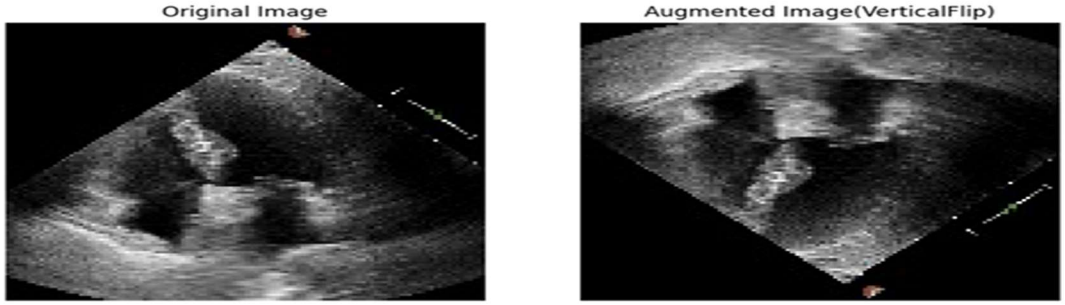


Figure 4. 4. Vertical Flipping

4 .2.3. Model Preparation

Model Summary (Supervised Learning)

```
----------------------------------------------------------------
        Layer (type)               Output Shape          Param #
================================================================
            Conv2d1          [1, 64, 112, 112]            9,408
       BatchNorm2d-2          [1, 64, 112, 112]              128
              ReLU-3          [1, 64, 112, 112]                0
         MaxPool2d-4            [1, 64, 56, 56]                0
            Conv2d-5            [1, 64, 56, 56]           36,864
       BatchNorm2d-6            [1, 64, 56, 56]              128
              ReLU-7            [1, 64, 56, 56]                0
            Conv2d-8            [1, 64, 56, 56]           36,864
       BatchNorm2d-9            [1, 64, 56, 56]              128
             ReLU10            [1, 64, 56, 56]                0
       BasicBlock11            [1, 64, 56, 56]                0
            Conv2d12            [1, 64, 56, 56]           36,864
      BatchNorm2d13            [1, 64, 56, 56]              128
             ReLU14            [1, 64, 56, 56]                0
           Conv2d15            [1, 64, 56, 56]           36,864
      BatchNorm2d16            [1, 64, 56, 56]              128
             ReLU17            [1, 64, 56, 56]                0
       BasicBlock18            [1, 64, 56, 56]                0
            Conv2d19           [1, 128, 28, 28]           73,728
      BatchNorm2d-20           [1, 128, 28, 28]              256
             ReLU-21           [1, 128, 28, 28]                0
           Conv2d-22           [1, 128, 28, 28]          147,456
      BatchNorm2d-23           [1, 128, 28, 28]              256
           Conv2d-24           [1, 128, 28, 28]            8,192
      BatchNorm2d-25           [1, 128, 28, 28]              256
             ReLU-26           [1, 128, 28, 28]                0
       BasicBlock-27           [1, 128, 28, 28]                0
           Conv2d-28           [1, 128, 28, 28]          147,456
      BatchNorm2d-29           [1, 128, 28, 28]              256
             ReLU-30           [1, 128, 28, 28]                0
           Conv2d-31           [1, 128, 28, 28]          147,456
      BatchNorm2d-32           [1, 128, 28, 28]              256
             ReLU-33           [1, 128, 28, 28]                0
       BasicBlock-34           [1, 128, 28, 28]                0
           Conv2d-35           [1, 256, 14, 14]          294,912
      BatchNorm2d-36           [1, 256, 14, 14]              512
             ReLU-37           [1, 256, 14, 14]                0
           Conv2d-38           [1, 256, 14, 14]          589,824
      BatchNorm2d-39           [1, 256, 14, 14]              512
           Conv2d-40           [1, 256, 14, 14]           32,768
      BatchNorm2d-41           [1, 256, 14, 14]              512
             ReLU-42           [1, 256, 14, 14]                0
       BasicBlock-43           [1, 256, 14, 14]                0
           Conv2d-44           [1, 256, 14, 14]          589,824
      BatchNorm2d-45           [1, 256, 14, 14]              512
             ReLU-46           [1, 256, 14, 14]                0
           Conv2d-47           [1, 256, 14, 14]          589,824
      BatchNorm2d-48           [1, 256, 14, 14]              512
             ReLU-49           [1, 256, 14, 14]                0
       BasicBlock-50           [1, 256, 14, 14]                0
           Conv2d-51             [1, 512, 7, 7]        1,179,648
      BatchNorm2d-52             [1, 512, 7, 7]            1,024
             ReLU-53             [1, 512, 7, 7]                0
           Conv2d-54             [1, 512, 7, 7]        2,359,296
      BatchNorm2d-55             [1, 512, 7, 7]            1,024
           Conv2d-56             [1, 512, 7, 7]          131,072
      BatchNorm2d-57             [1, 512, 7, 7]            1,024
             ReLU-58             [1, 512, 7, 7]                0
       BasicBlock-59             [1, 512, 7, 7]                0
           Conv2d-60             [1, 512, 7, 7]        2,359,296
      BatchNorm2d-61             [1, 512, 7, 7]            1,024
             ReLU-62             [1, 512, 7, 7]                0
           Conv2d-63             [1, 512, 7, 7]        2,359,296
      BatchNorm2d-64             [1, 512, 7, 7]            1,024
             ReLU-65             [1, 512, 7, 7]                0
       BasicBlock-66             [1, 512, 7, 7]                0
           Linear-67                   [1, 512]       12,845,568
             ReLU-68                   [1, 512]                0
          Dropout-69                   [1, 512]                0
          Encoder-70                   [1, 512]                0
           Linear-71                   [1, 256]          131,328
             ReLU-72                   [1, 256]                0
          Dropout-73                   [1, 256]                0
           Linear-74                   [1, 128]           32,896
             ReLU-75                   [1, 128]                0
          Dropout-76                   [1, 128]                0
           Linear-77                   [1, 256]           33,024
  ProjectionHead-78                   [1, 256]                0
================================================================
Total params: 24,219,328
Trainable params: 24,219,328
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.57
Forward/backward pass size (MB): 62.81
Params size (MB): 92.39
Estimated Total Size (MB): 155.77
----------------------------------------------------------------
```

Figure 4. 5. Supervised Learning Model Summary

Segmentation Model

```
Segmentation Model Summary:
==========================================================================================
Layer (type:depth-idx)                             Output Shape              Param #
==========================================================================================
UNet                                               [1, 1, 224, 224]          672
├─ResNet: 1-1                                      [1, 512]                  --
│    └─Conv2d: 2-1                                 [1, 64, 112, 112]         9,408
│    └─BatchNorm2d: 2-2                            [1, 64, 112, 112]         128
│    └─ReLU: 2-3                                   [1, 64, 112, 112]         --
│    └─MaxPool2d: 2-4                              [1, 64, 56, 56]           --
│    └─Sequential: 2-5                             [1, 64, 56, 56]           --
│    │    └─BasicBlock: 3-1                        [1, 64, 56, 56]           73,984
│    │    └─BasicBlock: 3-2                        [1, 64, 56, 56]           73,984
│    └─Sequential: 2-6                             [1, 128, 28, 28]          --
│    │    └─BasicBlock: 3-3                        [1, 128, 28, 28]          230,144
│    │    └─BasicBlock: 3-4                        [1, 128, 28, 28]          295,424
│    └─Sequential: 2-7                             [1, 256, 14, 14]          --
│    │    └─BasicBlock: 3-5                        [1, 256, 14, 14]          919,040
│    │    └─BasicBlock: 3-6                        [1, 256, 14, 14]          1,180,672
│    └─Sequential: 2-8                             [1, 512, 7, 7]            --
│    │    └─BasicBlock: 3-7                        [1, 512, 7, 7]            3,673,088
│    │    └─BasicBlock: 3-8                        [1, 512, 7, 7]            4,720,640
│    └─AdaptiveAvgPool2d: 2-9                      [1, 512, 1, 1]            --
│    └─Identity: 2-10                              [1, 512]                  --
├─ConvTranspose2d: 1-2                             [1, 256, 2, 2]            524,544
├─Sequential: 1-3                                  [1, 128, 2, 2]            --
│    └─Conv2d: 2-11                                [1, 256, 2, 2]            590,080
│    └─ReLU: 2-12                                  [1, 256, 2, 2]            --
│    └─BatchNorm2d: 2-13                           [1, 256, 2, 2]            512
│    └─Conv2d: 2-14                                [1, 128, 2, 2]            295,040
│    └─ReLU: 2-15                                  [1, 128, 2, 2]            --
│    └─BatchNorm2d: 2-16                           [1, 128, 2, 2]            256
├─ConvTranspose2d: 1-4                             [1, 64, 4, 4]             32,832
├─Sequential: 1-5                                  [1, 32, 4, 4]             --
│    └─Conv2d: 2-17                                [1, 64, 4, 4]             36,928
│    └─ReLU: 2-18                                  [1, 64, 4, 4]             --
│    └─BatchNorm2d: 2-19                           [1, 64, 4, 4]             128
│    └─Conv2d: 2-20                                [1, 32, 4, 4]             18,464
│    └─ReLU: 2-21                                  [1, 32, 4, 4]             --
│    └─BatchNorm2d: 2-22                           [1, 32, 4, 4]             64
├─ConvTranspose2d: 1-6                             [1, 16, 8, 8]             2,064
├─Sequential: 1-7                                  [1, 1, 8, 8]              --
│    └─Conv2d: 2-23                                [1, 16, 8, 8]             2,320
│    └─ReLU: 2-24                                  [1, 16, 8, 8]             --
│    └─BatchNorm2d: 2-25                           [1, 16, 8, 8]             32
│    └─Conv2d: 2-26                                [1, 1, 8, 8]              17
│    └─ReLU: 2-27                                  [1, 1, 8, 8]              --
│    └─BatchNorm2d: 2-28                           [1, 1, 8, 8]              2
├─Upsample: 1-8                                    [1, 1, 224, 224]          --
==========================================================================================
Total params: 12,680,467
Trainable params: 12,680,467
Non-trainable params: 0
Total mult-adds (Units.GIGABYTES): 1.82
==========================================================================================
Input size (MB): 0.60
Forward/backward pass size (MB): 39.83
Params size (MB): 50.72
Estimated Total Size (MB): 91.15
==========================================================================================
```

Figure 4. 6. Segmentation Model Summary

Model Train

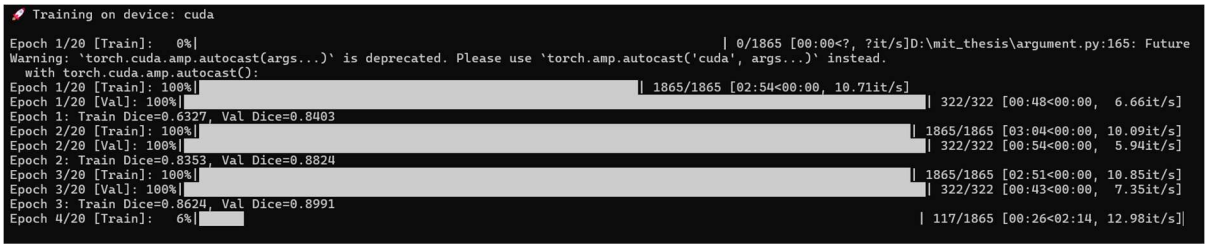


Figure 4. 7. Model Training

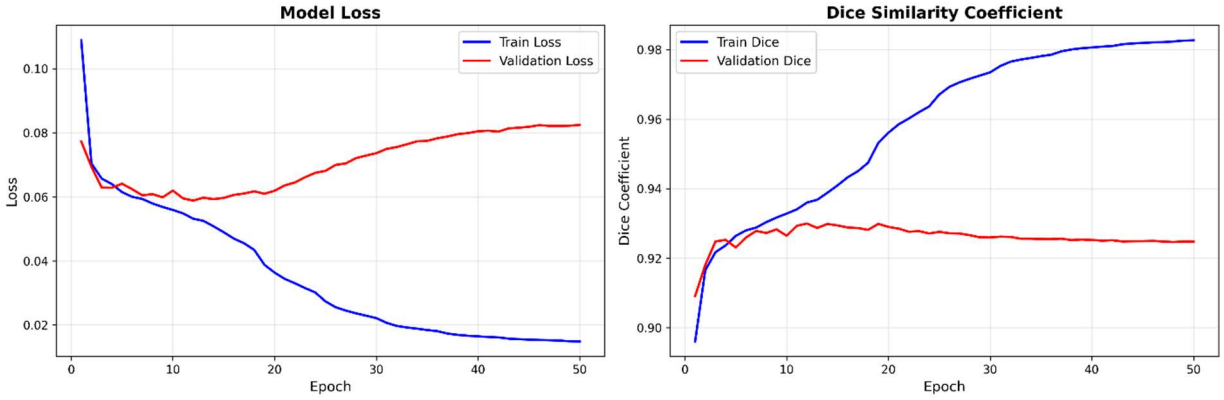


Figure 4. 8. Model Train without Data Augmentation with 50 Epoch

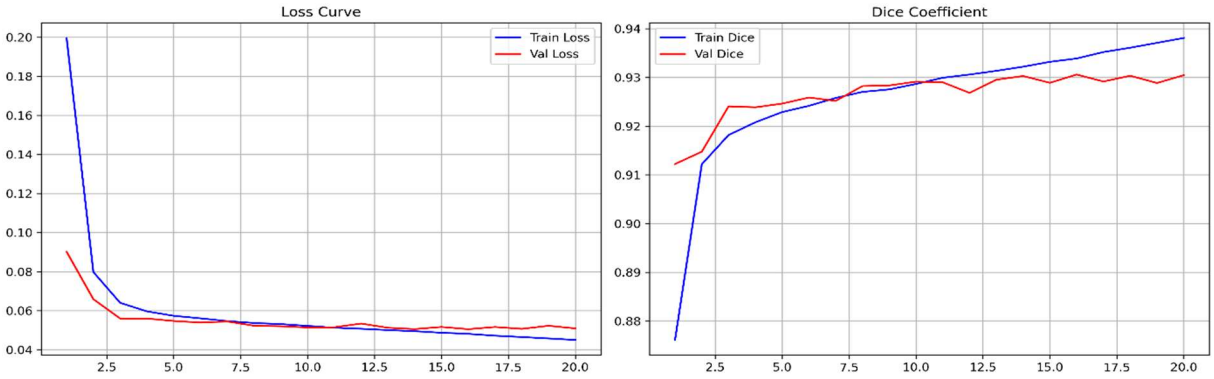


Figure 4. 9. Model Train without Data Augmentation with 20 Epoch

In both graphs, the training loss gradually drops, demonstrating the model's robust learning. Nevertheless, the validation loss plateaus and slightly rises in Figure 4.7 (50 epochs), indicating overfitting. While the validation Dice stabilizes at a lower value, indicating poor generalization, the training Dice coefficient rapidly improves. On the other hand, overfitting is lessened when training for fewer epochs (20 in Figure 4.8), although training and validation

measures stay closer. Overall, the model learns well, but without regularization or data augmentation, it has trouble generalizing. Techniques like data augmentation or model modification are required to improve validation performance and avoid overfitting.

Supervise Learning

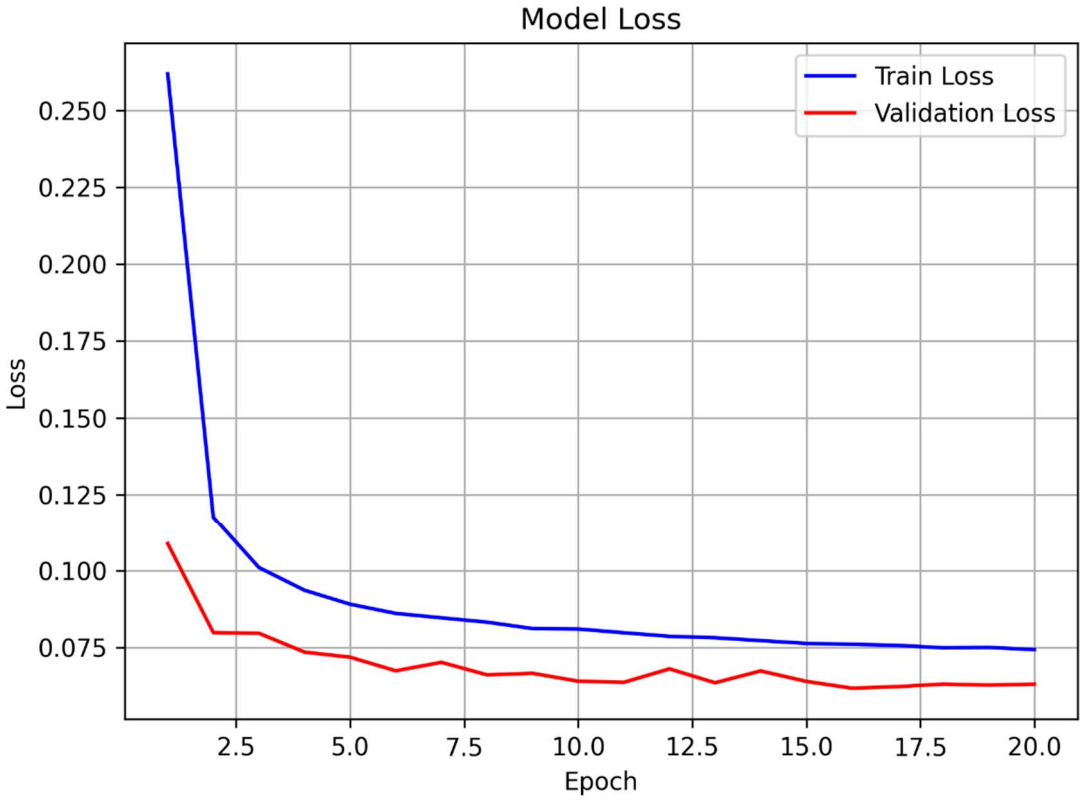


Figure 4. 10. Model train after Data Augmentation

The training and validation loss curves for the supervised learning model over 20 epochs are shown in Figure 4.9. Starting at about 0.26, the training loss quickly decreases throughout the first few epochs, reaching about 0.075 by the twentieth. In a similar vein, the validation loss starts at about 0.11, progressively declines, and stabilizes about below 0.07 as training comes to a close. The model is successfully learning and converging, as shown by the distinct decreasing trend in both loss curves. There is no indication of overfitting and good generalization since the validation loss continuously stays lower than the training loss. The model's embedding is becoming well-formed, enabling it to distinguish between positive and negative pairs more accurately, as evidenced by the loss values' gradual reduction and plateauing.

116,917 photos were used to train the supervised learning model, while 20,360 images were set aside for testing. A batch size of 64, 20 epochs, a learning rate of 0.001, the Adam optimizer, and a 50% dropout rate were among the hyper parameters used in the training process that were applied consistently to both training and testing datasets. The figure illustrates the close alignment between the training and validation loss curves, which was made possible by these parameters. The conclusion that the model training is dependable and successful is amply supported by the data trends. Overall, the findings support the model's effective learning and good generalization to fresh, untested data.

Accuracy Curve

Over the course of the training epochs, the accuracy and dice coefficient curves exhibit consistent progress. Training and validation Dice scores both rise, but the validation Dice score is consistently higher, suggesting that the model is doing a good job of generalizing. As training progresses, the model's improved segmentation performance is reflected in the Dice coefficient, which gauges the overlap between predicted and ground truth masks. Data augmentation, which offers a varied collection of training images, is responsible for this improvement by assisting the model in learning more resilient characteristics and preventing overfitting.

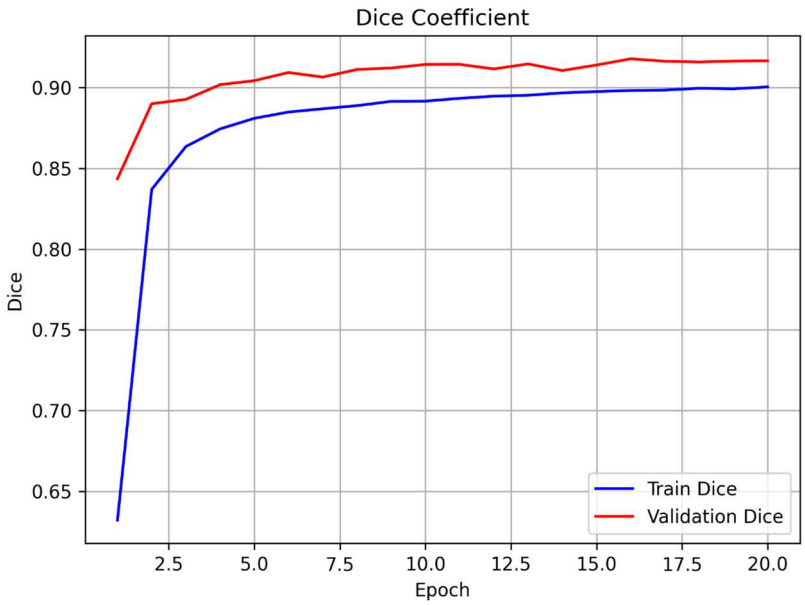


Figure 4. 11. Accuracy Curve

As the figure 4.1 show the accuracy curve for our model training. In the initial phase, for few epochs, both IoU and Dice-Coefficient are too low, as the model just begin to learn at initial stage of training. And as training continue, our accuracy improves as epochs increase as shown in figure there is a narrow, which means that our segmentation model used to perform better from the training data. At middle phase of model training, model predictions for segmentation align with the ground truths value. As we see in the figure, the upward trend becomes smoother. And at later phase of model training, the model learned most of the patterns from the training data. Which is good for our model training for segmentation of echocardiography image.

Segmentation:

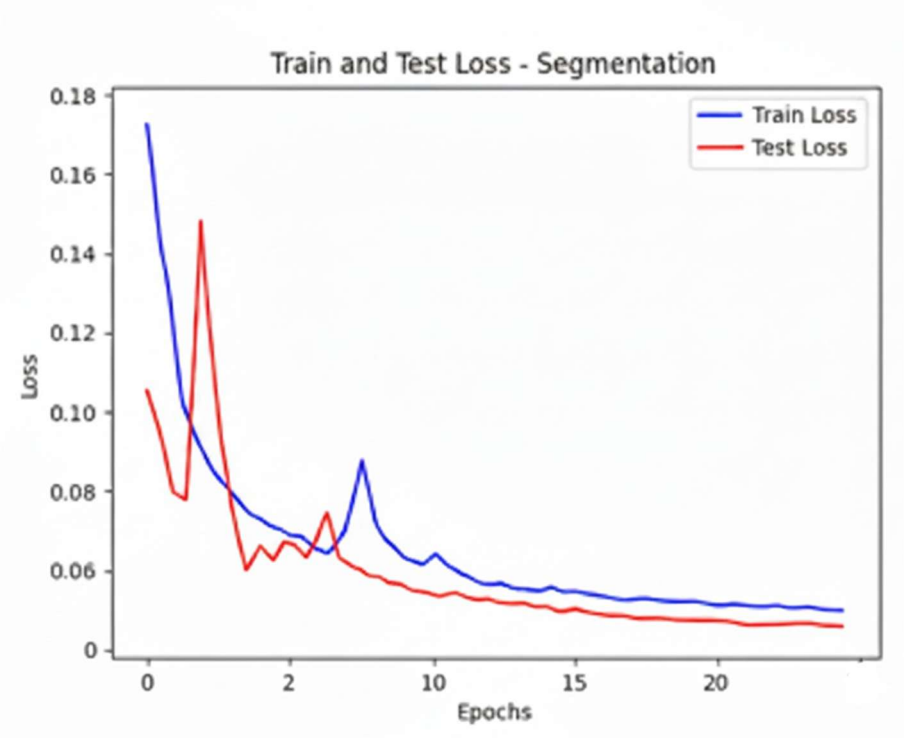


Figure 4. 12. Segmentation Model train

The segmentation model's training and test loss curves are displayed in Figure 4.11. The model was trained on 14,920 images and assessed on 2,552 test images. The model was trained for 20 epochs using BCE with Logits Loss as the loss function, a batch size of 64, and a learning rate of 0.001. Training and test losses both gradually decline, demonstrating the model's efficient convergence and good generalization to new data.

Predicted Output of Model:

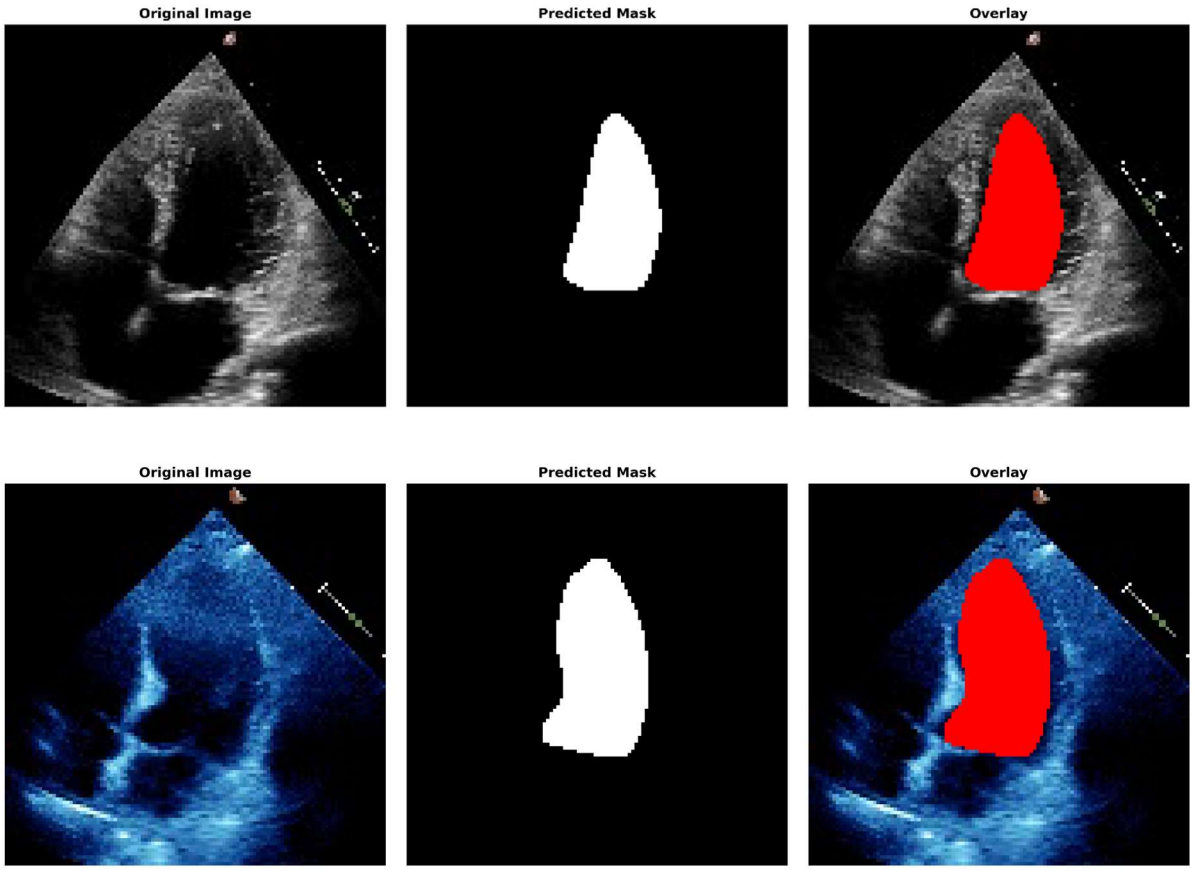


Figure 4. 13.Segmentation Model Output

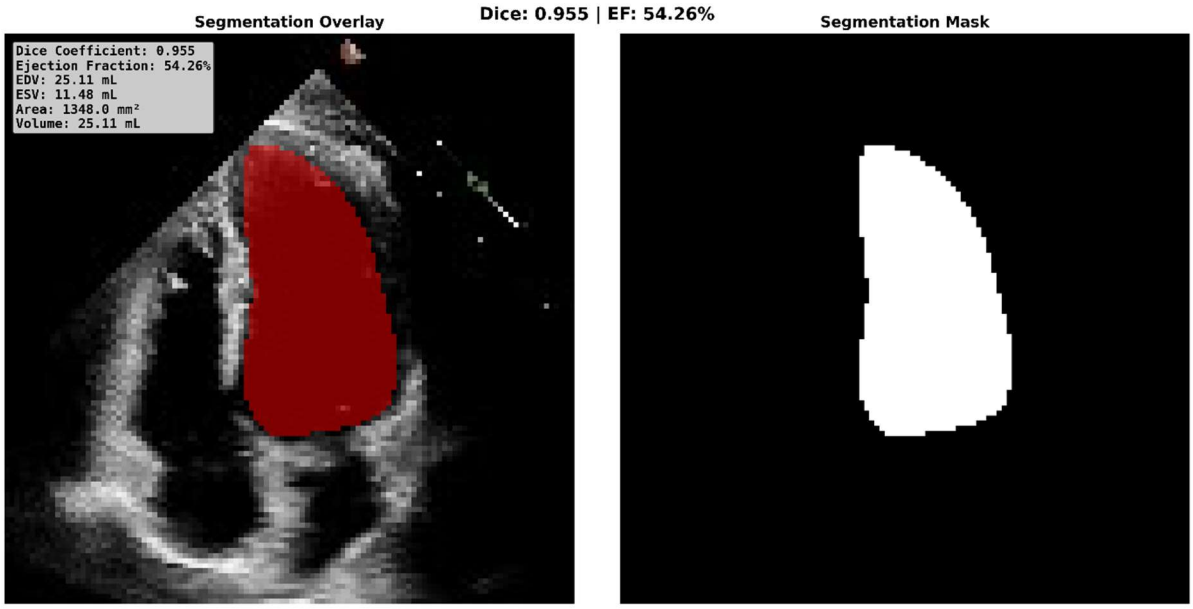
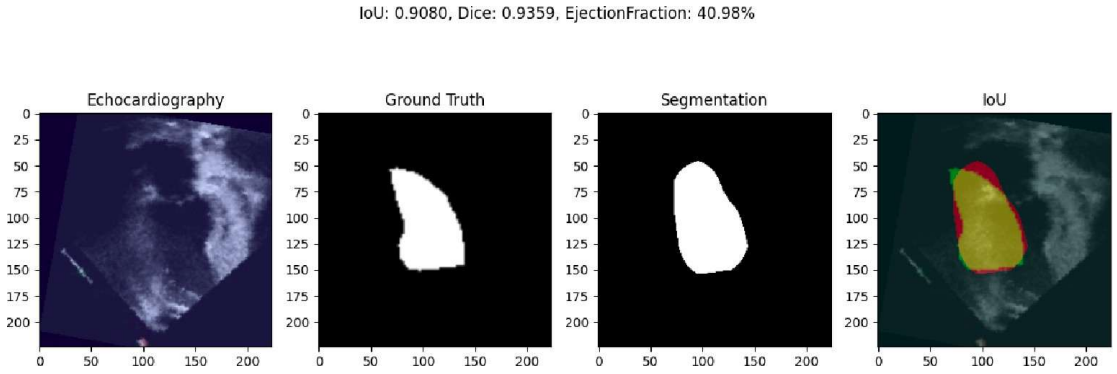


Figure 4. 14. Ejection Fraction Calculation

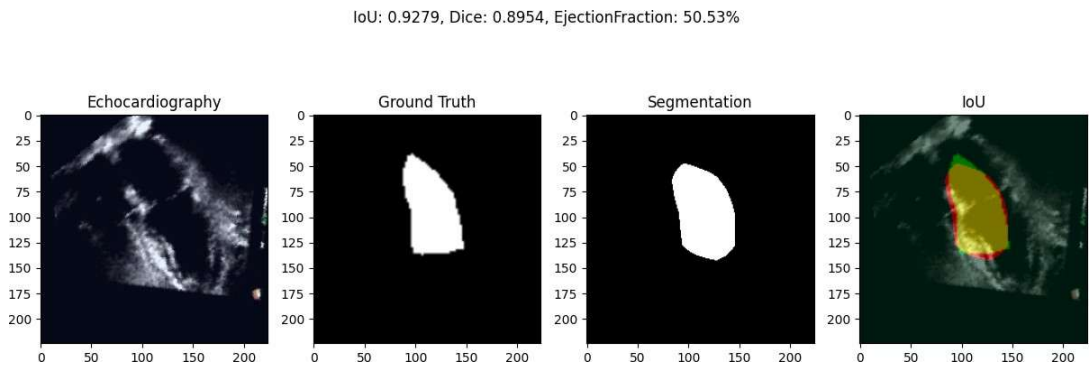
Unit 5: Result and Analysis

5.1. Finding and Observations

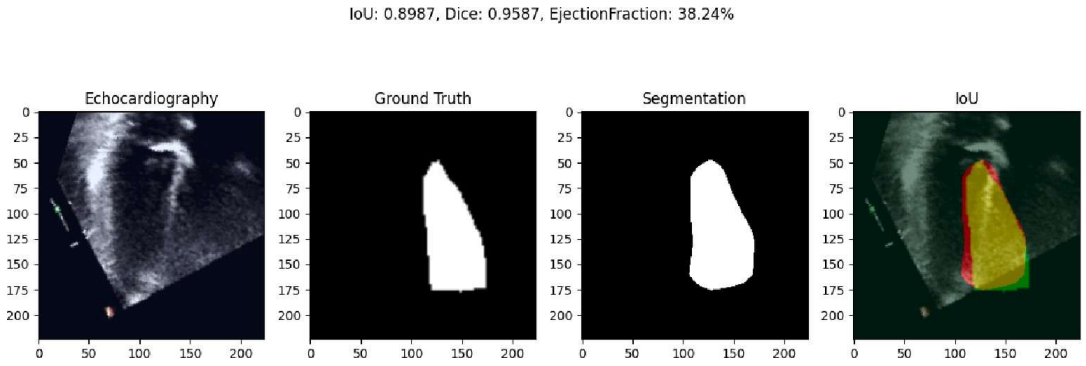
Here are some segmented outputs of the echocardiography containing their respective origin image, masked image, segmented output along with Calculated Intersection Over Union and Dice-coefficient and Ejection fraction for each image are shown as:



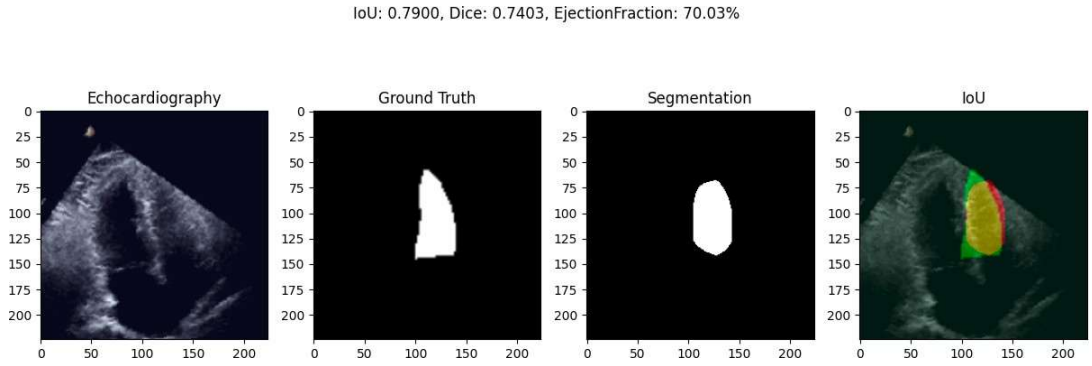
Here, in the first result, we can see that calculated ejection fraction for this specific heart is 40.98%. This ejection fraction lies in the range of Mild dysfunction, means that this heart is not immediate life-threatening, it suggests that the heart is not pumping as effectively as it should, but there is still sufficient output for normal activities. Heart in this range might have early stage of heart failure. Similarly calculated IoU is 90.80% and Dice coefficient is 93.59%.



In this result, it has calculated ejection fraction of 50.53%, which lies in the normal section, meaning the heart is pumping blood effectively and efficiently. Heart in this range do not have any major concerns related to heart failure. Also we get IoU of 92.79% and Dice coefficient of 89.54%.



Here we found ejection fraction rate of 38.24%, which lies in the range of Moderate dysfunction, this heart is significantly less efficient at pumping blood, which can lead to fluid buildup and symptoms of heart failure. This heart might be at risk of heart failure and this heart require medications or treatments to prevent further decline in heart function. And this heart has got IoU and Dice coefficient of 89.87% and 98.87% respectively.



Here, we got ejection fraction 70.03% for this specific heart. This is high ejection fraction rate; usually high ejection fraction might be seen in certain conditions such as hyperthyroidism or arteriovenous fistulas. Generally, it is not a cause of concern un- less associated with other symptoms or conditions. In this range of heart, pumps blood more effectively than usual, but this could be also reflecting abnormal condition in certain pathologies. And IoU 79% of and Dice coefficient of 74.03%.

And last result show ejection fraction rate of 21.97% which is in the Severe Dysfunction region, which means that heart is pumping less than 30% of its blood with each contraction.

This is critical condition of heart and cause heart failure. Heart in this range are at high risk for severe complications. Also, it also gets IoU of 79.70% and Dice coefficient of 94.82%.

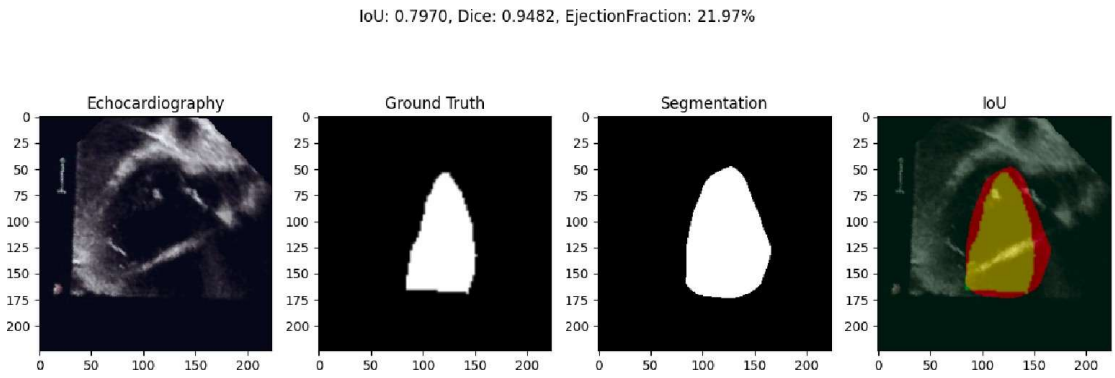


Figure 4. 15. Segmented Output with IoU, Dice Coefficient, and Ejection Fraction results.

As demonstrated by consistently high IoU and Dice coefficients, analysis of the sample echocardiograms, along with their related masks and segmented outputs, verifies that the suggested model is successful in recognizing cardiac structures. Ejection fraction (EF) computation was consistently supported by the segmentation quality, allowing functional evaluation under a variety of circumstances. The findings showed a range of cardiac function: mild dysfunction (40.98%), moderate dysfunction (38.24%), severe dysfunction (21.97%), and hyper dynamic cardiac function (70.03%). Overall, these results confirm the model's clinical value for automated cardiac evaluation as well as its robustness in generating accurate segmentation.

5.2. Result Analysis

5.1.1. Ejection Fraction

Table 5. 1.Ejection Fraction with problem

S.No.	Ejection Fraction (%)	Problem
1	40.98	Mild dysfunction
2	50.53	Normal
3	38.24	Moderate dysfunction
4	70.03	Hyperdynamic
5	21.97	Severe dysfunction

Table 5.1 shows the calculated ejection fraction of echocardiography images. And we categorize each ejection fraction rate if that specific heart has some problem or not. Normal means that heart is healthy. Similarly, Hyper dynamic indicate that heart need intense care of health support. All the standard ejection fraction measurement range by ACC with problem category are shown in Table 2. From the Table 4, it clearly sees that, Hearts image number 1(EF of 40.98%), 2(EF of 50.53%), 3(EF of 38.24%), 4(EF of 70.03%), 5(EF of 21.97%), are in Server dysfunction, Moderate dysfunction, Normal, Hyper dynamic and Mild dysfunction state. As result from Table 4, First result is in Mild Dysfunction state. This result is in slightly below normal, indicating some degree of heart weakness but often without significant symptoms. Second result show that hearts are in normal state. This range is recognized as healthy and demonstrates that the heart is circulating a proper amount of blood with each contraction. Third result is in Moderate Dysfunction state. A more noticeable reduction in heart function, likely lead- ing to symptoms like fatigue and shortness of breath during moderate physical activity. Fourth result indicate a very strong contraction of the heart, often seen in conditions where the heart is overworking. And last result shows the heart having Severe Dys- function problem state. This indicates severe weakening of the heart's pumping ability, often associated with significant symptoms even at rest or with minimal activity, and a higher risk of heart failure and other complications. This type of heart problem needs to be in intensive medical high care.

Model Performance

Table 5. 2.Model Performance

S.No.	IoU(%)	Dice coefficient (%)
1	0.9080	0.9353
2	0.9279	0.8954
3	0.8987	0.9587
4	0.7900	0.7403
5	0.7970	0.9482

Table 5.2 summarizes the segmentation model's performance, which was assessed using the Dice Coefficient and Intersection over Union (IoU) metrics. The model was trained over 20 epochs using 116,917 annotated echocardiography pictures with a batch size of 64, a scaling

factor (τ) of 0.1, and a learning rate of 0.001. IoU and Dice Coefficient both have a range of 0 to 1, with higher numbers denoting superior segmentation performance.

The model demonstrated good segmentation abilities for the evaluated configuration, achieving an IoU of 92.79% and a Dice Coefficient of 95.87% (Table 5.1). The model maintained an average IoU of 89.67% and an average Dice Coefficient of 91.02% over the whole test dataset of 2,552 pictures, demonstrating its strong generalization and dependable segmentation accuracy under various data augmentation conditions.

Comparison of results

Here are comparison results of our proposed method with other research work [10] and [11].

Unit 6: Conclusion and Future Recommendations

6.1. Conclusion

In order to overcome the problem of inadequate labelled data in medical imaging, our study creates a supervised learning-based model to compute the ejection fraction (EF) from echocardiography pictures. To acquire reliable feature representations, the encoder is initially trained on labelled pictures. These representations are subsequently fed into a U-Net backbone for precise cardiac segmentation. Reliable EF estimate requires accurate heart structure segmentation, and using a pre-trained encoder greatly improves model performance. The model obtained an average IoU of 89.67% and a Dice score of 91.02%, with top values of 92.79% and 95.87%, respectively, when evaluated using Intersection over Union (IoU) and Dice coefficient metrics.

These findings show that supervised learning reduces the need for laborious manual annotations and advances automated cardiac analysis by enabling effective, automated, and accurate EF computation.

6.2. Future Recommendation

Here are some future research plans:

• By modifying Supervised learning methods to effectively capture temporal and spatial dynamics in echocardiogram pictures, research can be improved.

• Future research could focus on developing explainable models that offer critical regions in the heart in addition to heart segmentation and ejection fraction predictions. Clinical decision-making would be enhanced by this.

• To capture temporal changes in heart architecture, extend supervised learning algorithms to analyses echocardiography video sequences instead of static frames.

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